

WATERTOWN, SD, USA

WMO: 726546

Lat: 44.905N

Lon: 97.149W

Elev: 1748

StdP: 13.79

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14946

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.3	-10.5	-20.7	1.9	-14.6	-16.6	2.4	-10.1	32.3	11.7	28.4	15.2	8.0	320	0.666

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.8	89.4	73.1	86.4	71.0	83.6	69.4	76.0	85.4	73.9	82.8	71.9	80.4	13.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.1	131.2	81.5	71.0	121.9	79.8	69.0	113.6	77.5	40.7	85.7	38.7	82.8	36.7	80.5	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
28.0	25.0	22.3	-22.8	95.1	6.2	3.1	-27.3	97.4	-30.9	99.2	-34.4	101.0	-38.9	103.3		
			-23.0	78.9	5.9	2.0	-27.2	80.3	-30.6	81.5	-33.9	82.6	-38.2	84.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.5	13.0	16.4	29.0	43.3	55.8	66.2	71.6	69.0	60.7	46.3	31.2	18.1
	DBStd	22.98	13.21	13.49	13.12	10.42	8.90	6.86	6.09	5.80	8.48	9.52	11.36	12.05
	HDD50	4796	1146	942	659	249	45	1	0	0	12	186	568	988
	HDD65	8388	1611	1362	1117	653	308	65	14	26	182	584	1013	1453
	CDD50	2431	0	0	7	47	225	488	670	587	332	70	5	0
	CDD65	547	0	0	0	1	22	102	219	149	51	3	0	0
	CDH74	5122	0	0	2	32	262	961	2089	1259	471	46	0	0
	CDH80	1571	0	0	0	5	64	274	729	366	125	8	0	0
(14) Wind	WSAvg	10.6	10.7	11.0	11.4	12.1	11.5	9.9	8.9	8.7	10.0	11.1	11.0	10.9
(15) Precipitation	PrecAvg	22.9	0.6	0.6	1.2	2.2	3.1	4.0	3.1	2.8	2.2	1.7	0.9	0.5
	PrecMax	40.0	2.1	1.9	3.9	6.5	9.7	11.4	11.5	6.4	5.8	8.1	3.1	2.4
	PrecMin	10.8	0.0	0.0	0.1	0.1	0.4	0.6	0.5	0.5	0.2	0.0	0.0	0.0
	PrecStd	5.7	0.5	0.5	0.9	1.4	2.1	2.4	2.0	1.6	1.4	1.7	0.7	0.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.8	51.7	68.7	78.2	86.4	91.2	94.7	91.5	88.5	80.4	65.6	47.6
		MCWB	36.7	44.4	53.5	58.8	65.8	72.4	75.2	74.0	69.7	61.3	51.9	41.3
	2%	DB	37.1	44.0	59.6	71.2	81.3	86.3	90.3	87.5	83.8	72.1	58.3	40.8
		MCWB	33.5	38.9	50.4	54.6	64.5	69.9	74.6	72.6	68.3	58.3	48.3	35.6
	5%	DB	34.1	38.3	53.2	65.8	76.4	83.0	87.0	84.0	79.9	66.2	51.8	36.8
		MCWB	32.0	35.1	46.4	51.2	61.4	68.2	72.4	70.1	65.9	55.2	44.5	33.4
	10%	DB	31.2	34.7	46.9	60.7	71.8	79.7	84.0	81.0	75.5	61.4	46.6	33.5
		MCWB	29.5	32.3	40.9	48.8	58.5	66.1	70.8	68.6	64.1	53.1	40.4	31.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.4	46.1	56.8	60.5	69.7	75.7	79.7	77.8	73.5	65.3	54.9	41.5
		MCDB	42.5	50.8	62.7	74.5	81.6	86.8	88.4	87.3	82.0	75.3	63.2	46.6
	2%	WB	34.1	39.0	51.3	56.2	66.3	73.0	77.0	74.9	70.9	60.4	49.5	36.4
		MCDB	36.4	43.1	58.2	68.2	77.5	82.8	86.4	83.8	79.8	68.3	56.7	40.0
	5%	WB	32.1	35.3	46.4	52.8	63.5	70.5	74.9	72.7	68.8	57.0	44.7	33.6
		MCDB	34.1	38.2	53.5	62.7	73.7	80.2	83.9	80.7	77.0	65.1	51.6	36.4
	10%	WB	29.5	32.3	41.1	50.0	60.4	68.3	72.8	70.7	66.2	53.4	40.9	31.5
		MCDB	31.1	34.7	46.9	59.4	69.3	77.2	81.5	78.2	73.3	60.5	46.3	33.6
(35) Mean Daily Temperature Range	5% DB	MDBR	16.9	17.9	17.6	21.1	20.6	19.9	20.8	20.5	21.4	19.5	17.4	15.4
		MCDBR	19.3	20.1	26.2	30.1	27.1	23.9	23.4	23.4	25.6	27.7	25.9	19.8
		MCWBR	16.5	15.7	17.1	16.5	14.3	11.9	10.8	11.4	12.5	16.1	16.7	15.5
	5% WB	MCDBR	18.6	18.8	25.8	26.3	23.4	20.7	21.0	21.8	24.4	25.1	18.8	
		MCWBR	16.1	14.7	17.3	15.5	13.6	11.8	11.1	11.4	12.4	15.2	16.7	15.3
(40) Clear-Sky Solar Irradiance	taub		0.262	0.282	0.306	0.352	0.369	0.379	0.384	0.390	0.350	0.314	0.283	0.261
	taud		2.429	2.396	2.429	2.333	2.367	2.387	2.370	2.337	2.430	2.504	2.497	2.448
	Ebn at Noon		276	289	296	288	285	282	279	273	276	274	266	265
	Edh at Noon		22	28	31	38	38	37	37	37	31	25	21	20
(44) All-Sky Solar Radiation	RadAvg		573	912	1204	1534	1771	1982	2111	1754	1366	887	589	430
	RadStd		61	93	116	132	129	129	78	146	98	111	58	40

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page